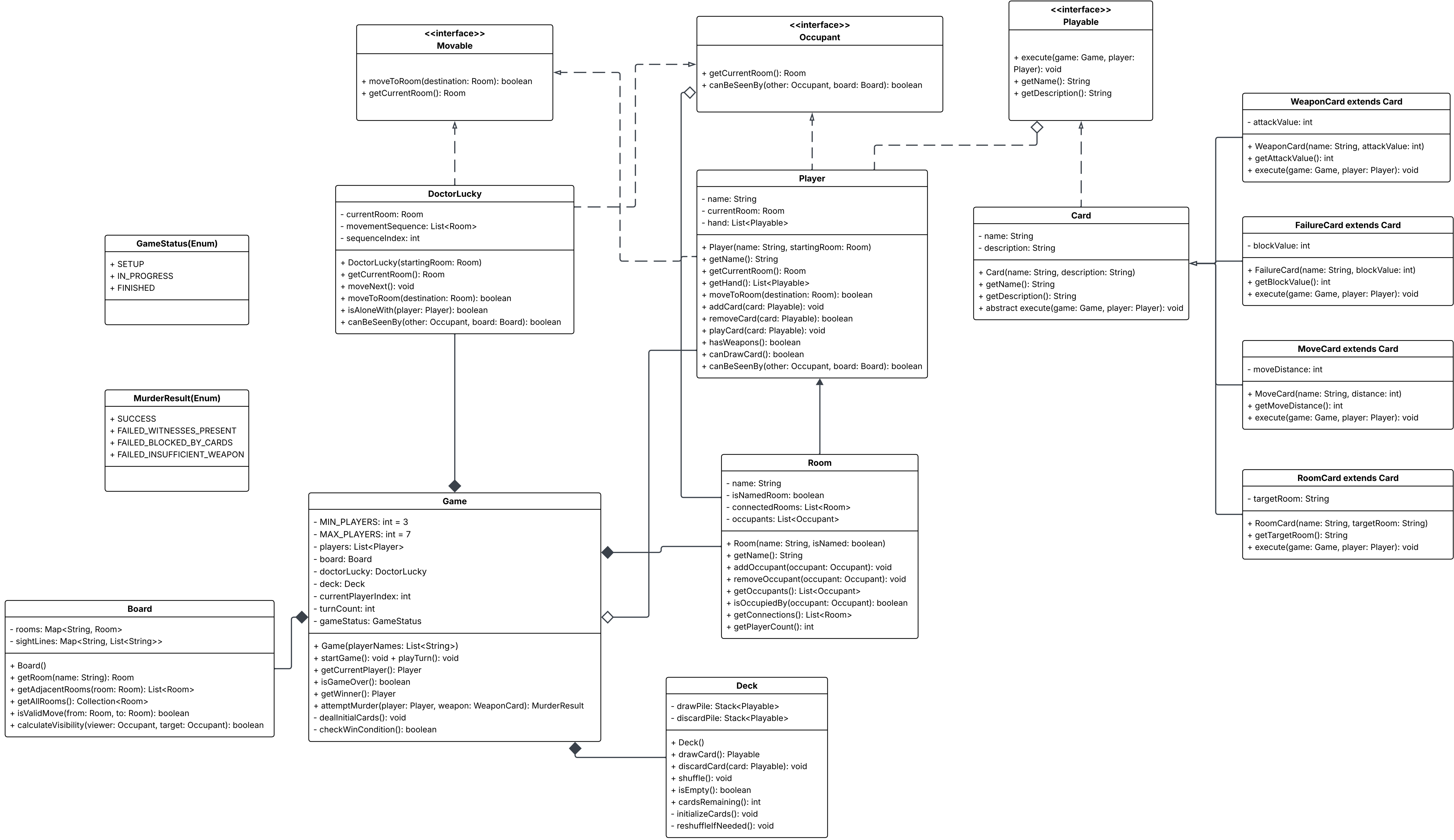




Test Plan for KillDoctorLucky

- Game Class Test
 - Valid game initialization
 - Condition: Game constructor with valid player names
 - Example Data: ["Peter", "Will", "Alice", "Henry"]
 - Expected: the Game created, gameStatus = SETUP
 - Minimum Players(Valid)
 - Condition: Game constructor with minimum players to be 3
 - Example Data: ["Peter", "Will", "Alice"]
 - Expected: Game created successfully.
 - Maximum Players (Valid)
 - Condition: Game constructor with maxium players to be 7
 - Example Data: ["Peter", "Will", "Alice", "Henry", "Alan", "Olivia", "George"]
 - Expected: Game created successfully.
 - Below Minimum Players
 - Condition: Game constructor with 2 players
 - Example: ["Peter", "Henry"]
 - Expected: Exception thrown
 - Above Maximum Players
 - Condition:Game constructor with 8 players
 - Example:["Peter", "Will", "Alice", "Henry", "Alan", "Olivia", "George", "Jason"]
 - Expected: Exception thrown
 - Start Game
 - Condition: startGame called
 - Example: Game with 3 players
 - Expected: gameStatus changes from SETUP to IN_PROGRESS
 - Successful Murder
 - Condition: attemptMurder() with player alone, strong weapon
 - Example: Alice alone with Doctor Lucky, WeaponCard(value = 10)
 - Expected: MurderResult.SUCCESS, game over.
 - Failed Murder - Witness
 - Condition: attempMurder() with witness present
 - Example: Alice, Henry and Doctor Lucky all in same room
 - Expected: MurderResult.FAILED_WITNESS_PRESENT
- Player Class Test
 - Player Creation
 - Condition: Player constructor with valid data
 - Example: name = "Alice", startingRoom = Kitchen
 - Expected: Player created, currentRoom = Kitchen, empty hand
 - Valid Movement
 - Condition: moveToRoom() to connected room
 - Example: Player in Kitchen, move to Dining Room
 - Expected: Returns true, currentRoom = Dining Room
 - Invalid Movement
 - Condition: moveToRoom() to unconnected room
 - Example: Player in Kitchen, move to Bedroom (not connected)
 - Expected: Return False, currentRoom unchanged
 - Add Card to Hand
 - Condition: addCard() with space available
 - Example: Player with 3 cards, add WeaponCard("Knife", 5)
 - Expected: hand. size() = 4, card added
 - Hands initialization
 - Condition: Player starts with 6 cards
 - Example: New Game initialization
 - Expected: each player's hand.size() = 6
- Board and Room Test
 - Get Valid Room
 - Condition: getRoom() with existing room name
 - Example: getRoom("Kitchen")
 - Expected: Returns Kitchen room object
 - Invalid Room
 - Condition: getRoom() with non-existent room
 - Example: getRoom("InvalidRoom")
 - Expected: Returns null or throws exception
 - Add Occupant to Room
 - Condition: addOccupant() with player
 - Example: add Henry to Kitchen
 - Expected: Henry in Kitchen's occupant list
 - Room Visibility
 - Condition: calculateVisibility() between rooms
 - Example: Alice in Kitchen and Bob in Dining Room (connected)
 - Expected: Returns True (can see each other)
- Doctor Lucky Test
 - Sequential Movement
 - Condition: moveNext() called multiple times
 - Example: Movement sequence [Kitchen, Dining, Living], call 3 times
 - Expected: Kitchen to Dining to Living to Kitchen (circle)
 - Alone Check-True
 - Condition: isAloneWith() with only player present
 - Example: Kitchen has only Alice and Doctor Lucky
 - Expected: doctorLucky.isAloneWith() return True
 - Alone Check-False
 - Condition: isAloneWith() with multiple player
 - Example: Kitchen has Alice, Henry and Doctor Lucky
 - Expected: doctorLucky.isAloneWith() return False
- Card Test
 - Weapon Card
 - Condition: WeaponCard creation and execution
 - Example: WeaponCard("Knife", 5)
 - Expected: Card created attackValue = 5
 - Failure Card
 - Condition: FailureCard blocks murder attempt
 - Example: FailureCard("Bad Luck", 3) used during murder
 - Expected: reduce murder success by 3 points
 - Move Card
 - Condition: MoveCard allows extra movement
 - Example: MoveCard("Quick Step", 2)
 - Expected: Player can move up to 2 additional rooms
 - Room Card
 - Condition: RoomCard instant teleportation
 - Example: RoomCard("Kitchen Card", "Kitchen")
 - Expected: player instantly move to Kitchen
- Deck Test
 - Empty Deck Reshuffle
 - Condition: drawCard() when draw pile empty
 - Example: Empty draw pile, 20 cards in discard
 - Expected: discard reshuffled, card drawn
 - Draw Card
 - Condition: drawCard() with cards available
 - Example: Deck with 50 cards
 - Expected: return card, deck size decrease
 - Discard Card
 - Condition: discardCard() adds to discard pile
 - Example: discard WeaponCard("Knife", 5)
 - Expected: Card added to discard pile
- Interface Test
 - Playable Interface
 - Condition: All cards implement Playable
 - Example: WeaponCard, MoveCard treated as Playable
 - Expected: execute(), getName() work for all cards
 - Movable Interface
 - Condition: Player and Doctor Lucky implement Movable
 - Example: Both objects as Movable type
 - Expected: moveToRoom(), getCurrentRoom() work for both
- Integration Test
 - Game end
 - Condition: Game ends when murder successful
 - Example: Alice successfully murders Doctor Lucky
 - Expected: gameStatus = FINISHED, Alice declared winner
 - Multiple player interaction
 - Condition: players interact during murder attempt
 - Example: Alice attempts murder, Bob plays FailureCard
 - Expected: Bob's card affects Alice's murder attempt
 - Complete Turn
 - Condition: full player turn with movement and card play
 - Example: Alice moves, plays MoveCard, attempts murder
 - Expected: all actions execute correctly, turn passes to next player